

PBPK evaluation report — Alfentanil

Fit is acceptable (overall GMFE 1.524); parameters were fitted by the numerical optimizer.

Objective: Build a whole-body PBPK model for alfentanil that reproduces the observed plasma concentration-time data after intravenous and oral dosing across the studies provided. You decide the model structure and any parameters you cannot obtain from the given data, and you justify them. No modeling strategy and no reference results are provided. Overall GMFE 1.524 (reference 1.447); within-2-fold 86.6%. Structure: Cellular partition coefficient method - Rodgers and Rowland, Cellular permeability - PK-Sim Standard; processes CYP3A4.

Model, parameters & rationale

MODEL CHOICE

A whole-body PBPK structure with Rodgers-and-Rowland tissue partitioning is appropriate for alfentanil: it is a basic, moderately lipophilic small molecule (MW 416.5, pKa~6.5) bound mainly to alpha-1-acid glycoprotein rather than albumin, and Rodgers-Rowland explicitly accounts for ionized-base partitioning into acidic phospholipids as well as neutral-lipid/plasma-protein partitioning, which is mechanistically the right choice for this compound class. Cellular permeability was calculated with the PK-Sim standard (charge/MW/lipophilicity-based) method rather than measured Caco-2/PAMPA input, which is a reasonable default for a passively permeating, non-P-gp substrate, but it means the permeability term is a structural surrogate rather than an independently anchored value. Hepatic CYP3A4-mediated intrinsic clearance was implemented as the sole elimination pathway, consistent with the known biology (extensive CYP3A4 metabolism, <1% renal excretion of parent), so GFR fraction was fixed to zero rather than estimated. Oral dosing was handled with a transcellular intestinal permeability term to capture absorption rate, since alfentanil is highly soluble (992 mg/L) and absorption is expected to be permeability- rather than dissolution-limited; no active transport process was added given the absence of P-gp substrate evidence.

PARAMETERS

Lipophilicity = 1.4920107663657072 [estimated] sens=0.68 ref=1.8463211883
Permeability = 0.00037520157796026095 cm/min [estimated] sens=0.025 ref=0.0068752756625
Intrinsic clearance = 0.41670803288507974 l/min [estimated] sens=1 ref=0.5272297928
Specific intestinal permeability (transcellular) = 3.8376090327163534e-05 cm/min [estimated] sens=0.026 ref=0.00057373577138
Fraction unbound (plasma, reference value) = 0.1 [fixed] ref=0.1
Solubility at reference pH = 992.0 [fixed] ref=None
GFR fraction = 0.0 [fixed] ref=None
Reference pH = 6.5 [fixed] ref=None
Molecular weight = 416.52 [fixed] ref=None

PHARMACOLOGICAL RATIONALE

Intrinsic clearance (0.417 L/min, sensitivity 1.0) is the dominant estimated parameter and is essentially fully determined by the IV concentration-time data (no independent literature CL_{int} value was provided for comparison); given its sensitivity of 1.0, the fitted value can be trusted as a data-driven estimate of net CYP3A4 hepatic clearance, but it is a lumped/apparent term that will also absorb any misspecification in distribution or absorption if those are wrong. Lipophilicity was fitted to 1.49, sensitivity 0.681 (fairly well-informed by the data), but this is a clear departure from the literature logD(7.4) of 2.1-2.2 (Baneyx 2014/Jansson 2008). Because sensitivity is moderately high, this is not a case that can be dismissed as "within a wide plausible range" - the optimizer is using an effective lipophilicity appreciably lower than the measured value, most plausibly to compensate for the Rodgers-Rowland-predicted tissue distribution/V_{ss} or for correlation with the fixed fraction unbound (0.1, itself within the reported 8.6-12% range and consistent with literature), and this trade-off should be flagged as a structural/parameter-correlation artifact rather than evidence that the compound's true lipophilicity is lower than measured. Cellular permeability (3.75e-4 cm/min, sensitivity 0.025) and specific intestinal transcellular permeability (3.84e-5 cm/min, sensitivity 0.026) are both very weakly identified by this dataset - the fits are essentially insensitive to their values, so these numbers should be treated as largely unconstrained/default-driven outputs of the PK-Sim standard method rather than confirmed

Model, parameters & rationale

physiological estimates; the visibly worse oral performance (78.2% within 2-fold vs 91.4% IV) is consistent with these low-sensitivity absorption parameters not being meaningfully calibrated by the available oral data. The fixed inputs (fraction unbound 0.1, solubility 992 mg/L at pH 6.5, reference pH 6.5, molecular weight 416.52) all match their literature anchors well and were reasonably left unadjusted.

Model equations (ODE system)

COMPARTMENTS / STATE VARIABLES

PK-Sim's standard human whole-body structure: ~15 perfused organs (lung, brain, heart, kidney, gut, spleen, pancreas, liver, stomach, muscle, skin, bone, adipose, gonads) plus arterial and venous blood pools. Each organ is further divided into plasma, blood-cell, interstitial and intracellular sub-compartments, so the solved system is one mass-balance ODE per sub-compartment (order $\sim 10^2$ states). The physiological volumes V and flows Q are fixed by the individual; only the drug terms below are model choices.

GOVERNING EQUATIONS

Perfused tissue (distribution):

$$V_t \cdot dC_t/dt = Q_t \cdot (C_{art} - C_t / (K_{p,t} / R_{bp}))$$

(Each non-eliminating organ t (brain, heart, muscle, skin, adipose, bone, ...): rate in from arterial blood at flow Q_t , rate out with the venous effluent in equilibrium with tissue via the tissue:plasma partition coefficient $K_{p,t}$ (R_{bp} = blood:plasma ratio).)

Venous blood pool:

$$V_{ven} \cdot dC_{ven}/dt = \sum_t Q_t \cdot C_t / (K_{p,t} / R_{bp}) - Q_{co} \cdot C_{ven} + R_{iv}(t)$$

(Collects the effluent of every organ; Q_{co} = cardiac output. IV doses enter here as an input rate $R_{iv}(t)$ (bolus or infusion).)

Arterial blood pool:

$$V_{art} \cdot dC_{art}/dt = Q_{co} \cdot (C_{lung} - C_{art})$$

(Blood is oxygenated/mixed through the lung and distributed to the organs.)

Permeability-limited exchange (cellular):

$$V_{cell} \cdot dC_{cell}/dt = P \cdot SA \cdot (f_u \cdot C_{int} - f_{u,cell} \cdot C_{cell} / K_{p,t})$$

(Where cellular permeability limits uptake, the intracellular space is a separate state exchanging with interstitium across the membrane (permeability $P \times$ surface area SA), driven by the unbound gradient.)

Liver (hepatic metabolism):

$$V_{liv} \cdot dC_{liv}/dt = Q_{liv} \cdot C_{art} + Q_{po} \cdot C_{po} - Q_{hv} \cdot C_{liv} / (K_{p,liv} / R_{bp}) - CL_{int} \cdot f_u \cdot C_{liv}$$

(The liver receives arterial + portal (Q_{po} , first-pass) inflow; CYP3A4 metabolism removes drug as a first-order intrinsic-clearance sink $CL_{int} \cdot f_u \cdot C_{liv}$ on the unbound concentration.)

Kidney (renal filtration):

$$V_{kid} \cdot dC_{kid}/dt = Q_{kid} \cdot (C_{art} - C_{kid} / (K_{p,kid} / R_{bp})) - GFR \cdot f_u \cdot C_{kid}$$

(Glomerular filtration removes unbound drug at rate $GFR \cdot f_u \cdot C_{kid}$. In THIS model this term is ZERO (GFR fraction = 0): renal excretion of unchanged drug is negligible, so the renal sink is switched off.)

Oral absorption (PO doses):

$$dA_{lum}/dt = -P_{int} \cdot SA_{int} \cdot C_{lum,u} ; \quad J_{abs} = P_{int} \cdot SA_{int} \cdot C_{lum,u} \rightarrow \text{gut wall} \rightarrow \text{portal vein} \rightarrow \text{liver}$$

(Dissolved drug in the intestinal lumen permeates the mucosa (specific intestinal permeability $P_{int} \times$ surface area), enters the gut wall, and drains via the portal vein to the liver (first-

Model equations (ODE system)

pass).)

TISSUE PARTITIONING

$K_{p,t}$ is not a fitted number but is computed per organ from tissue composition by the Rodgers and Rowland method – a function of effective lipophilicity, the unbound fraction f_u , drug ionization (pK_a) and each tissue's lipid / water / phospholipid / protein content.

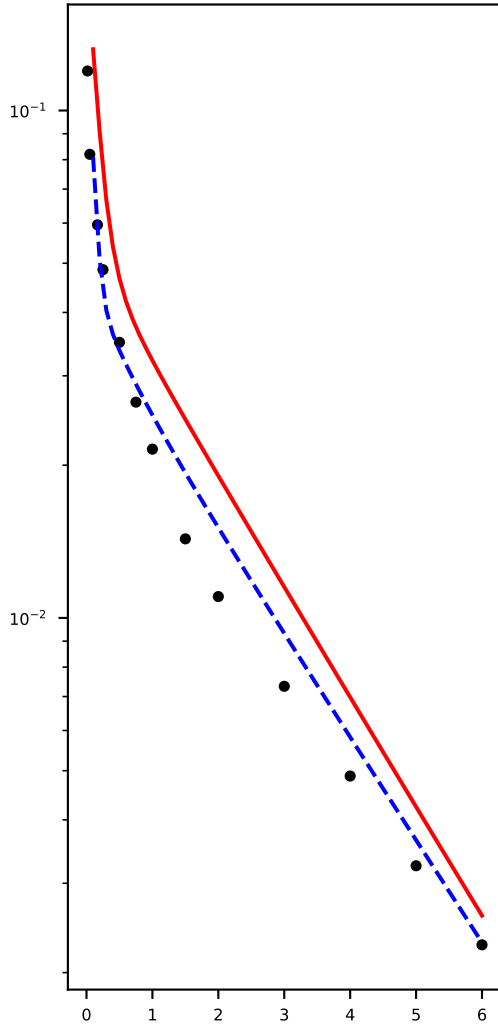
WHERE THE FITTED / FIXED PARAMETERS ENTER

- Lipophilicity: sets every tissue partition coefficient $K_{p,t}$ (the distribution terms in all organ ODEs).
- Permeability: the cellular permeability P in the permeability-limited exchange term.
- Intrinsic clearance: the hepatic elimination sink CL_{int} in the liver ODE.
- Specific intestinal permeability (transcellular): P_{int} in the oral-absorption flux J_{abs} .
- Fraction unbound (plasma, reference value) (fixed): f_u – scales every unbound-driven term ($K_{p,t}$, CL_{int} , GFR).
- GFR fraction (fixed): the renal filtration sink GFR in the kidney ODE.

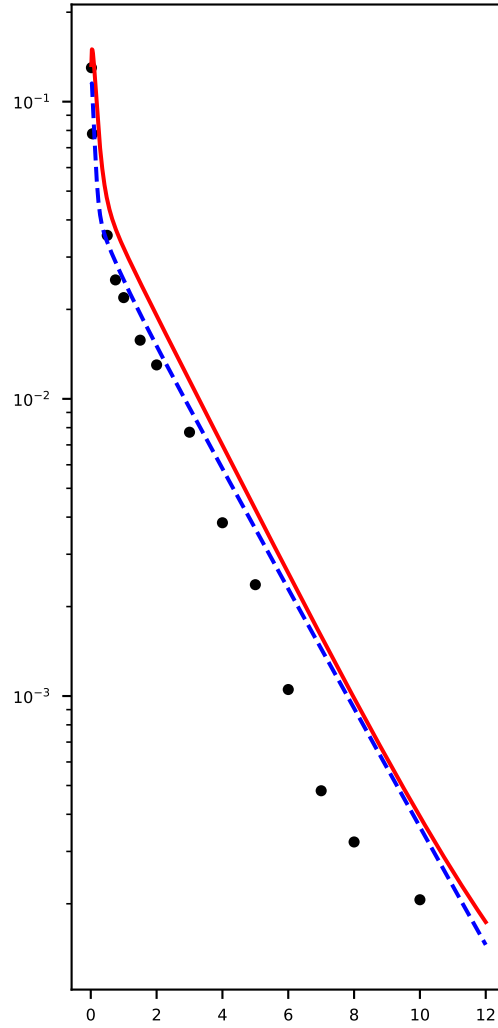
These are the governing equations; PK-Sim generates the fully instantiated system (every organ and sub-compartment, and the exact Rodgers and Rowland K_p formula) at build time and integrates it numerically. The concentrations plotted are the peripheral-venous-plasma state.

Observed (●) vs agent (—) vs reference (--)

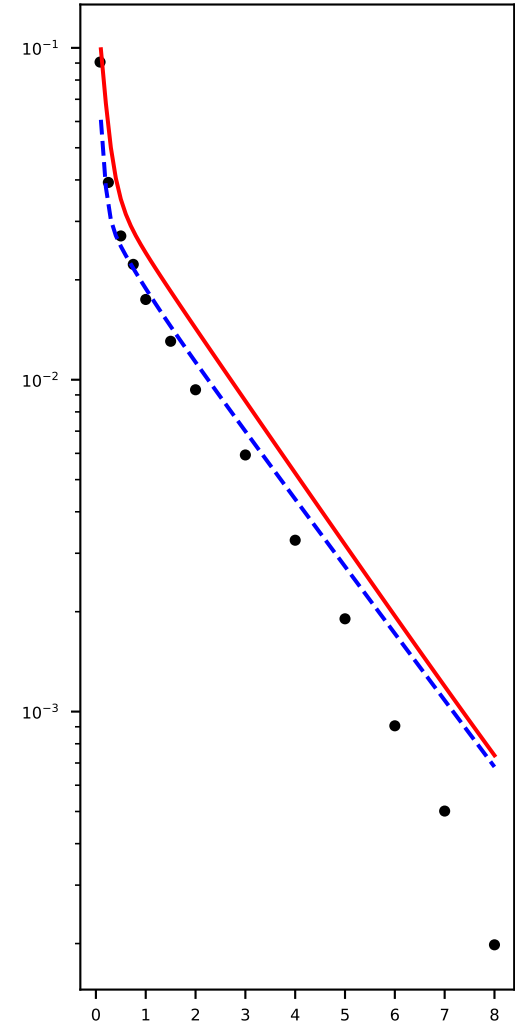
Meistelman_1987_Figure_1 adult



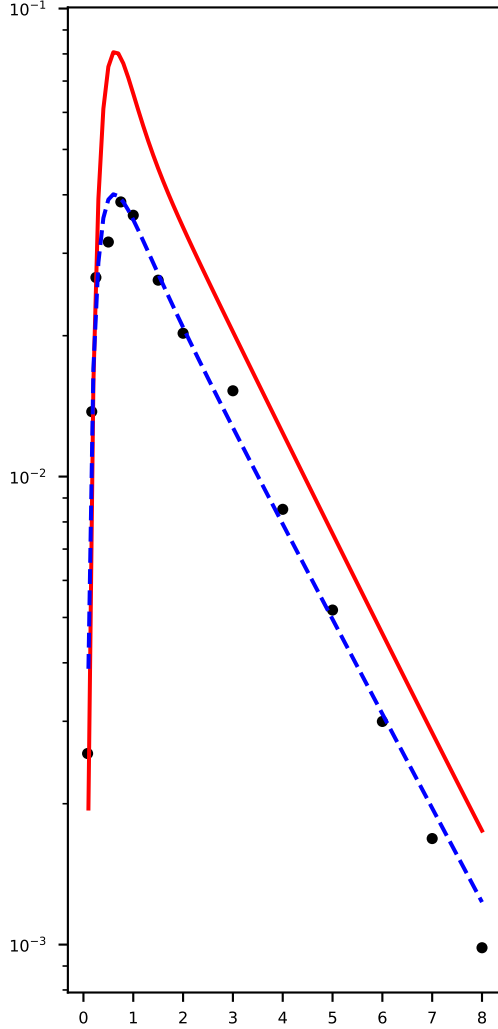
Kharasch 1997 - Control (Perpe



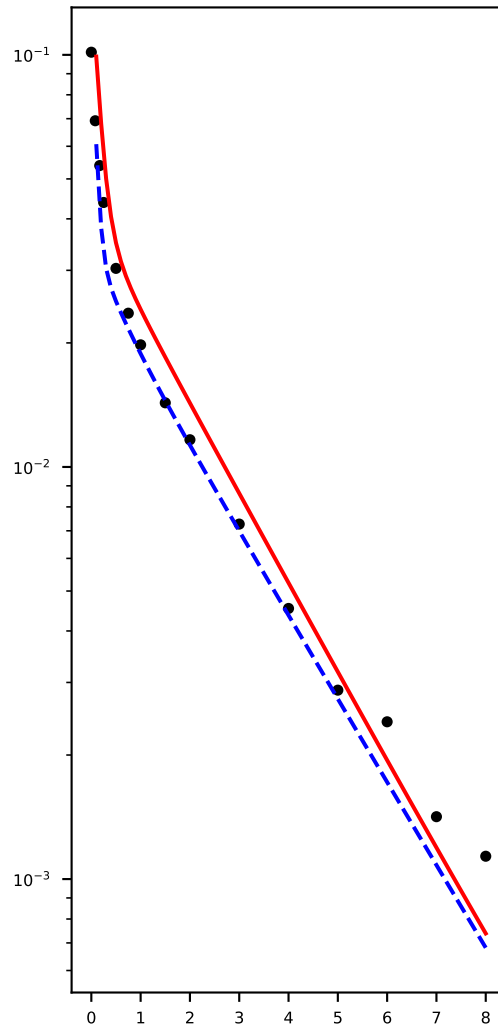
Kharasch 2004 - iv Control (Pe



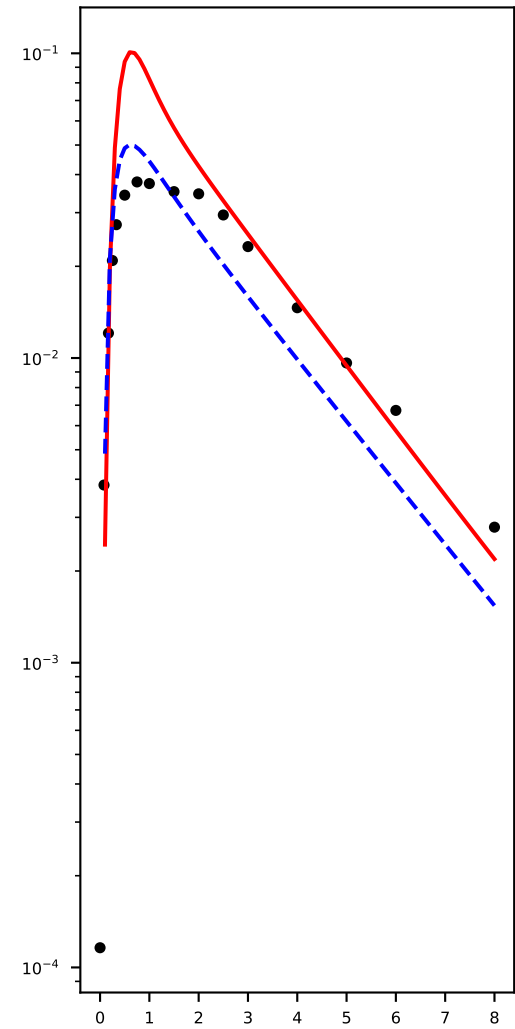
Kharasch 2004 - po #1 (60 μ g/k



Kharasch 2011 - iv Control (Pe

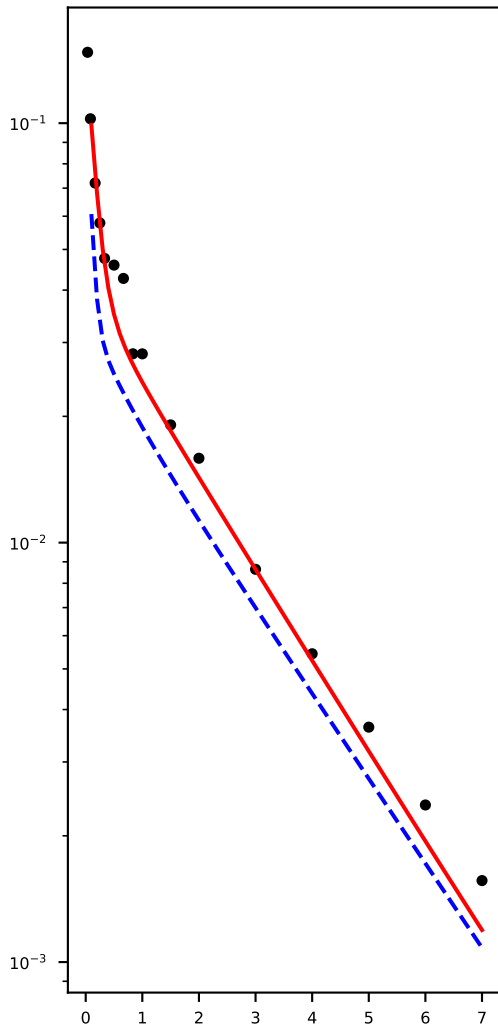


Kharasch 2011 - po Control (Pe

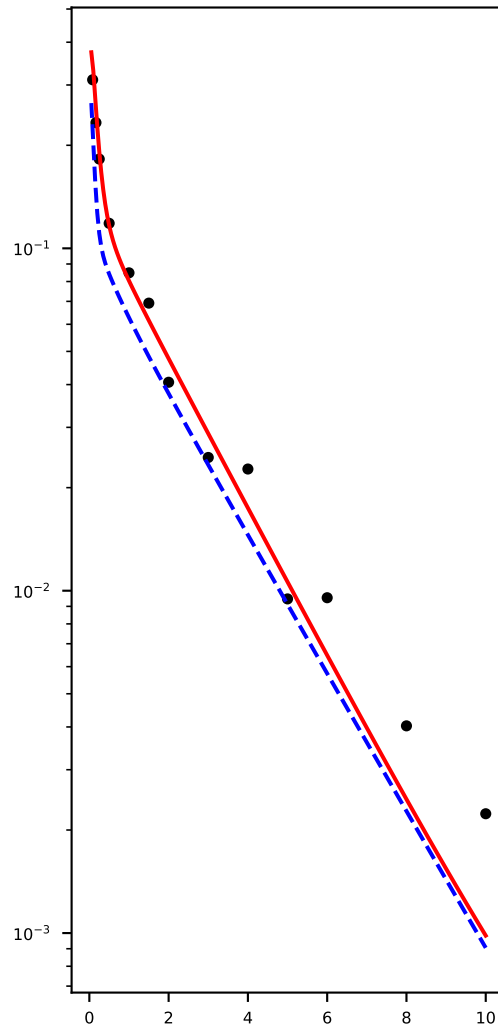


Observed (●) vs agent (—) vs reference (--)

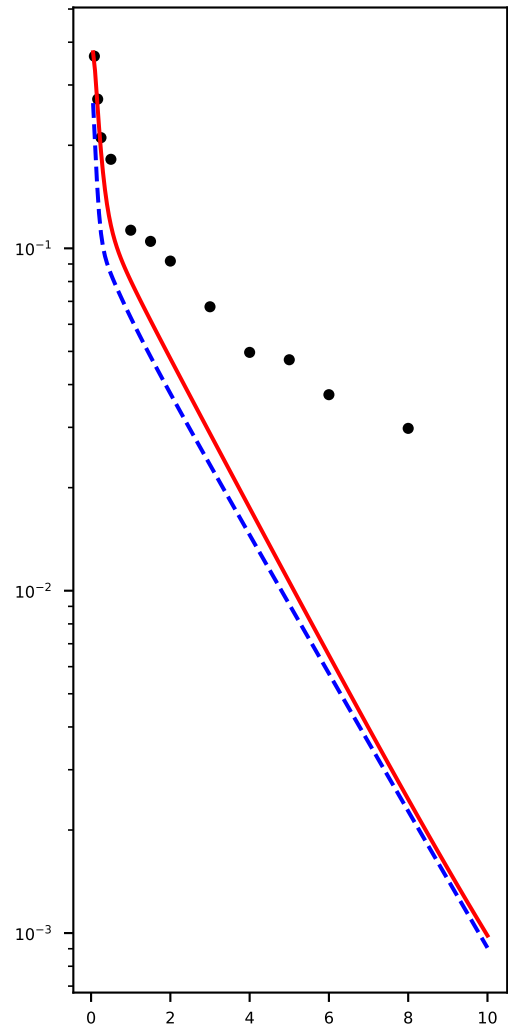
Phimmasone 2001 - Control (Per



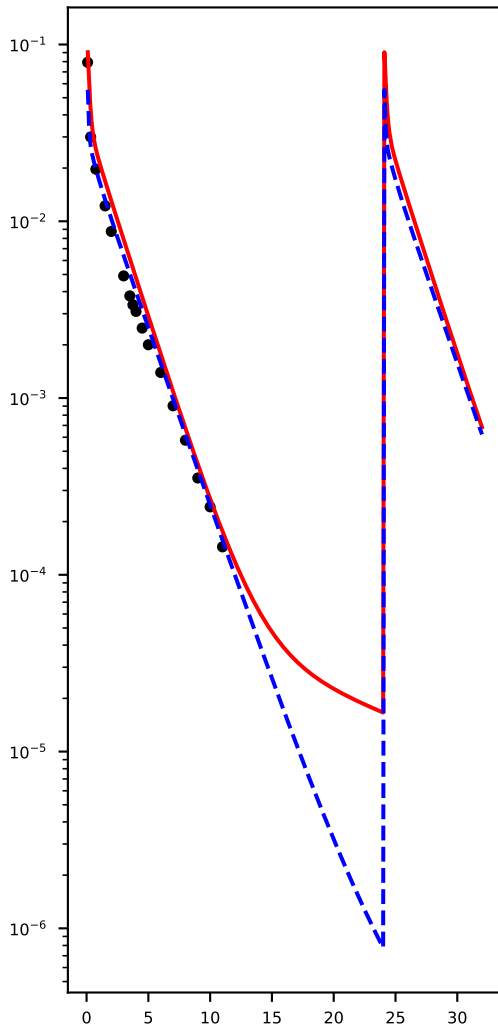
Ferrier 1985 - healthy control



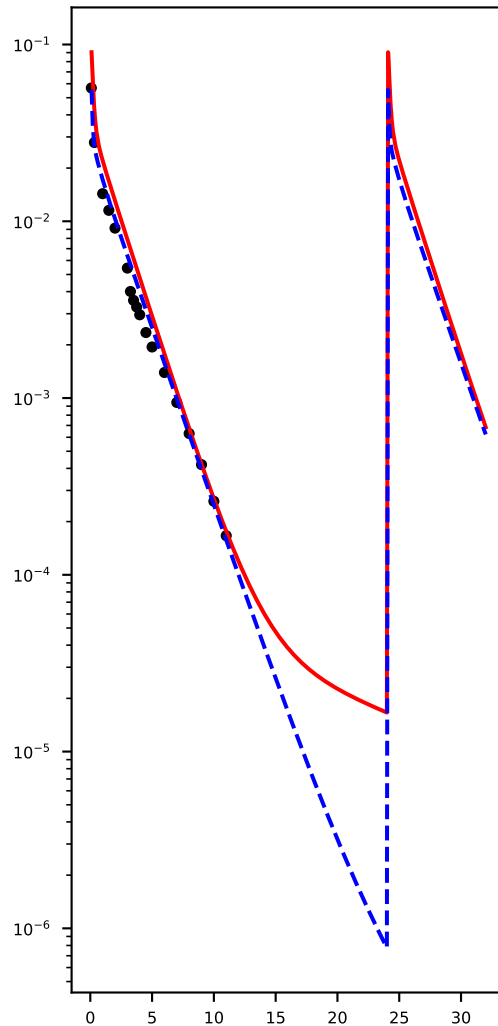
Ferrier 1985 - cirrhosis patie



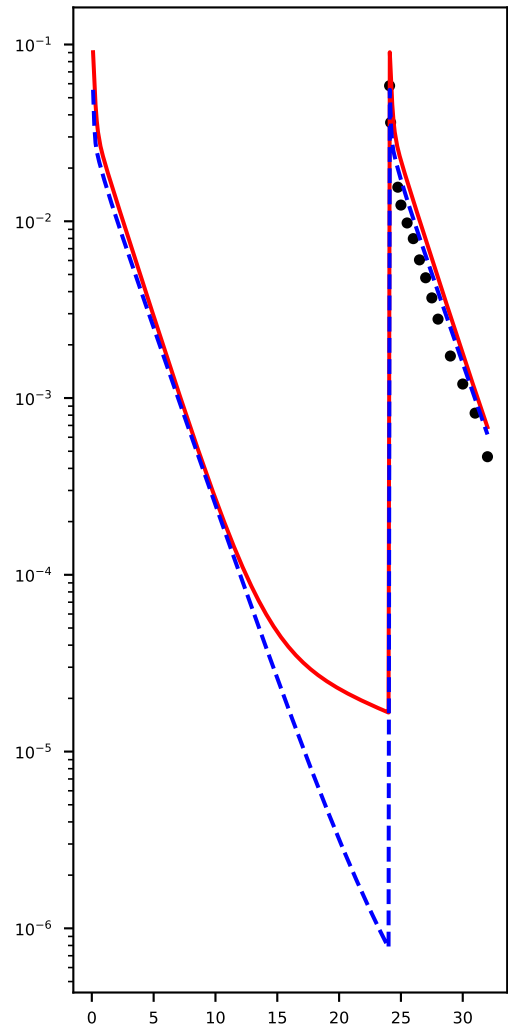
Kharasch 2011b - Alfentanil IV



Kharasch 2011b - Alfentanil IV

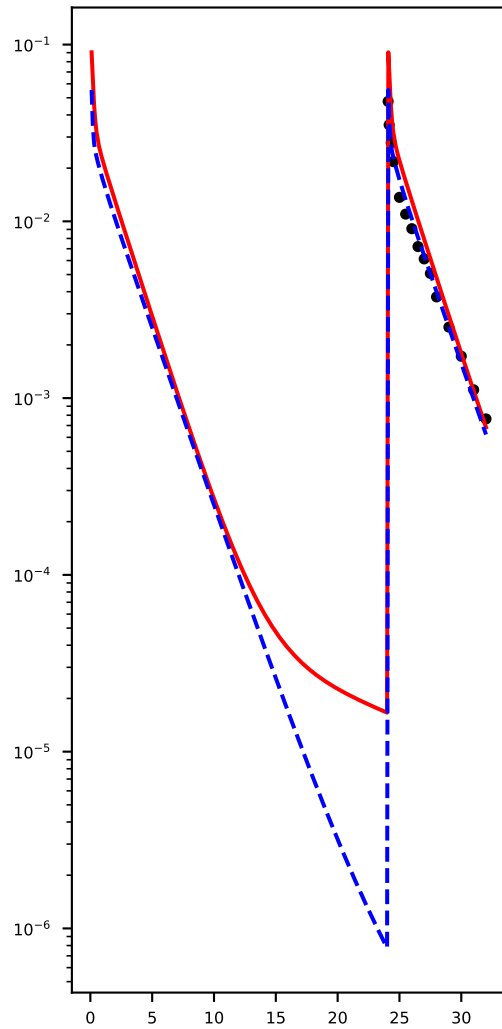


Kharasch 2011b - Alfentanil IV

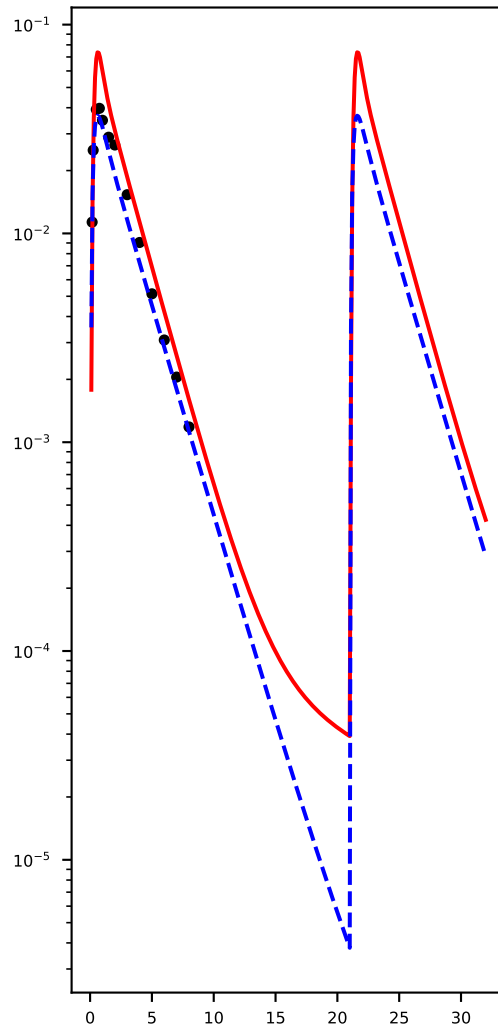


Observed (●) vs agent (—) vs reference (--)

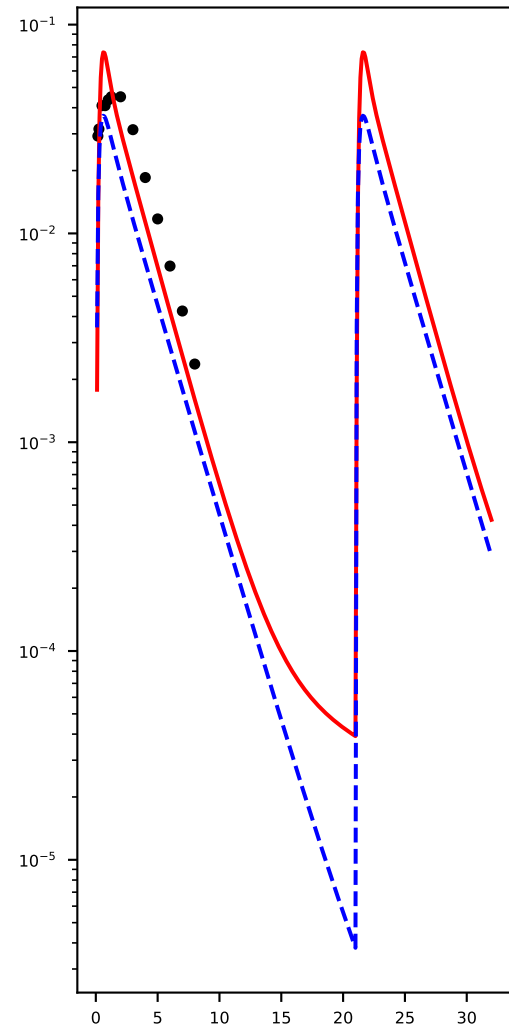
Kharasch 2011b - Alfentanil IV



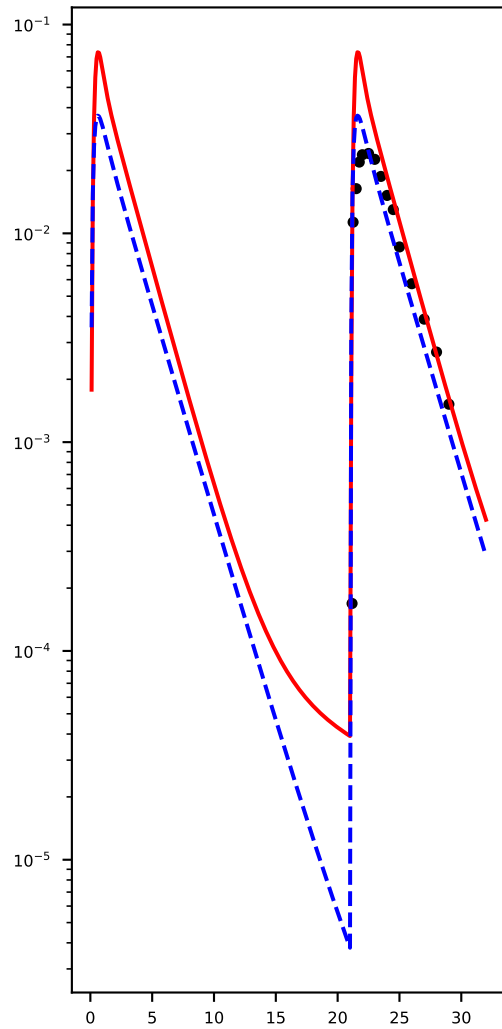
Kharasch 2011b - Alfentanil PO



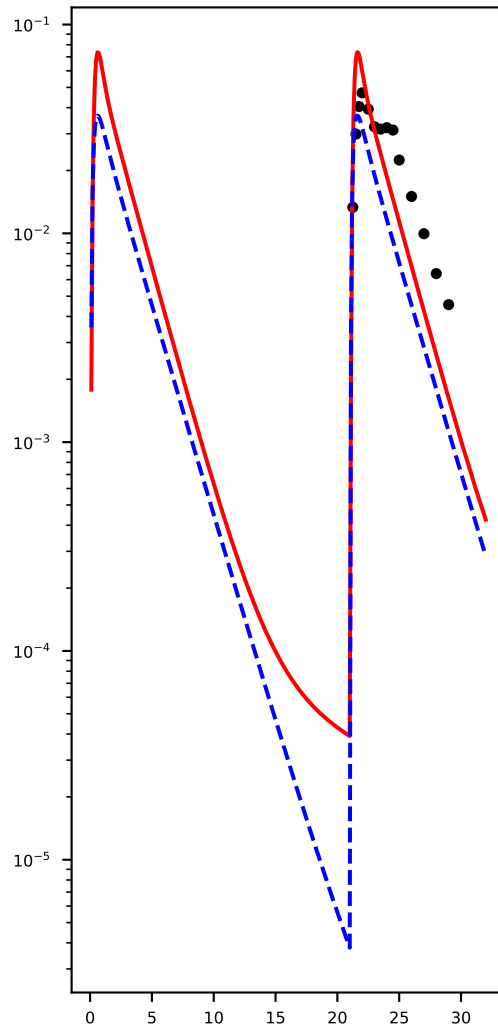
Kharasch 2011b - Alfentanil PO



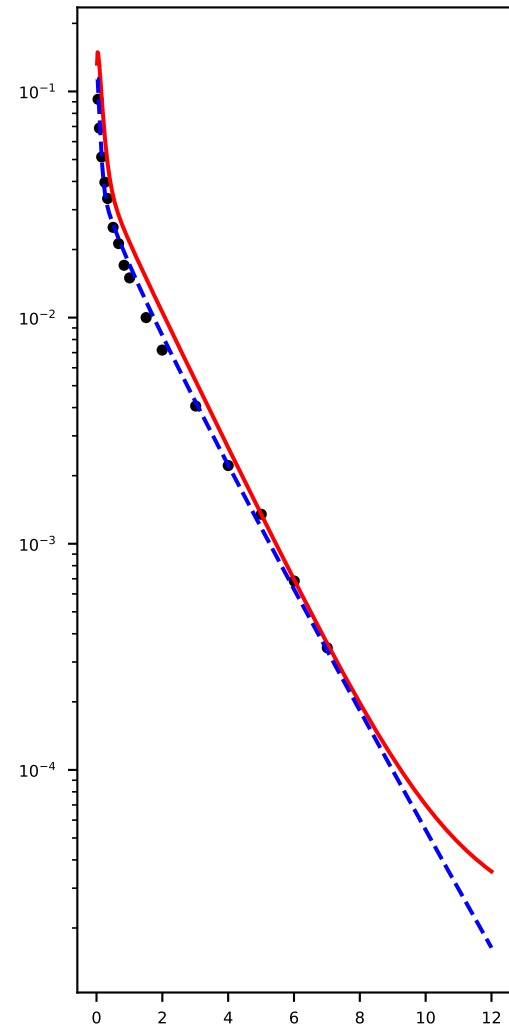
Kharasch 2011b - Alfentanil PO



Kharasch 2011b - Alfentanil PO

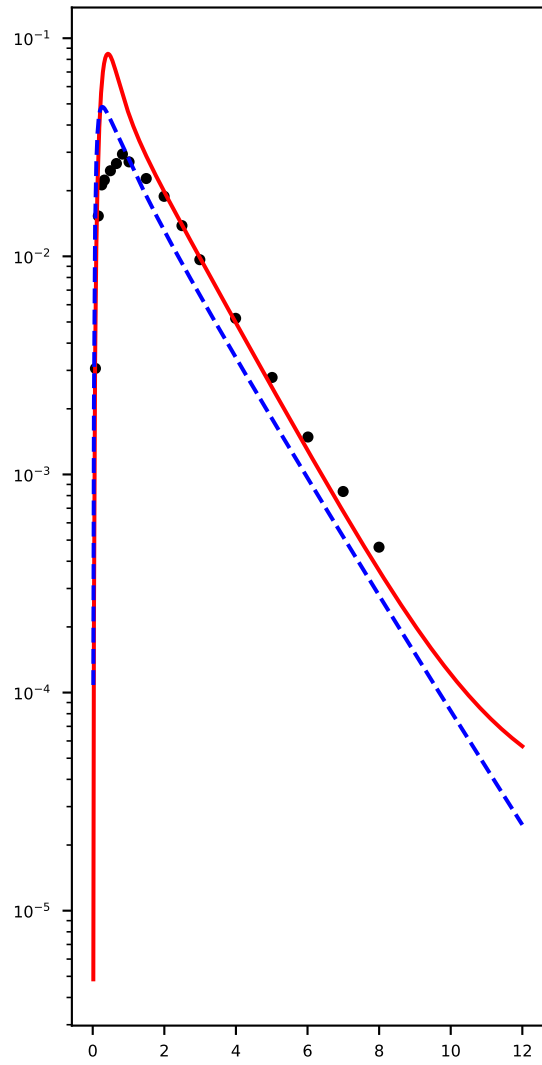


Kharasch2012_Alfentanil_alone_



Observed (●) vs agent (—) vs reference (--)

Kharasch2012_Alfentanil_alone_



Ground-truth comparison & conclusion

Post-hoc evaluation vs the reference model (not seen during modeling). Each parameter is judged on distance from truth AND identifiability - a value far from truth but weakly identified is not a real error.

Reference GMFE 1.447 vs this model 1.524

SUMMARY

The structure DIFFERS from the ground-truth model in: processes. Overall fit is comparable (this model GMFE 1.524 vs reference 1.447). Of 5 compared parameters, 3 recovered well, 2 differ but are weakly identified / minor, and 0 are off despite being influential.

STRUCTURE (this model | ground truth | match)

partition method: Rodgers and Rowland | Rodgers and Rowland | match
permeability method: PK-Sim Standard | PK-Sim Standard | match
processes: ['CYP3A4'] | ['CYP3A4', 'GlomerularFiltration'] | DIFFERS

PARAMETERS (this | truth | fold | sensitivity | assessment)

Lipophilicity: 1.492 | 1.846 | 0.81x | 0.68 | [GOOD] recovered well (0.81x)
Permeability: 0.0003752 | 0.006875 | 0.055x | 0.025 | [MINOR] far 0.055x but weakly identified - the data do not constrain it, so not a real error
Intrinsic clearance: 0.4167 | 0.5272 | 0.79x | 1 | [GOOD] recovered well (0.79x)
Specific intestinal permeability (transcellular): 3.838e-05 | 0.0005737 | 0.067x | 0.026 | [MINOR] far 0.067x but weakly identified - the data do not constrain it, so not a real error
Fraction unbound (plasma, reference value): 0.1 | 0.1 | 1x | - | [GOOD] fixed, matches the reference value (1x)
Solubility at reference pH: 992 | None | - | - | no reference value
GFR fraction: 0 | None | - | - | no reference value
Reference pH: 6.5 | None | - | - | no reference value
Molecular weight: 416.5 | None | - | - | no reference value

CONCLUSION

Optimization completed and parameters were fitted, yielding an overall fit rated "acceptable" (GMFE 1.524, 86.6% of observations within 2-fold, bias 1.214 indicating a modest systematic overprediction tendency). Performance is route-dependent: IV predictions are reasonably good (GMFE 1.467, 91.4% within 2-fold) while oral predictions are distinctly weaker (GMFE 1.628, 78.2% within 2-fold, larger bias), pointing to unresolved uncertainty in the absorption description. The high-sensitivity intrinsic clearance estimate is the most credible fitted parameter, well determined by the IV data, whereas the permeability terms (cellular and intestinal) are poorly constrained (sensitivity ~0.025-0.026) and should not be over-interpreted despite their biologically plausible magnitudes. The fitted lipophilicity, despite reasonably high sensitivity, disagrees with the measured logD and should be treated as an effective, correlated parameter rather than a validated physicochemical estimate; this departure, combined with the oral shortfall, suggests the distribution/absorption compensation in the model is not fully resolved. Overall the model is usable for IV alfentanil exposure but should be considered provisional for oral predictions and refined lipophilicity/absorption characterization before extrapolation to new dosing scenarios or populations.

Full modeling trajectory (LLM)

[Step 1] osp_inspect, osp_options

-> task description: objective, known biology, literature priors, current model, and the observed clinical data | authoritative action space: what you may edit and legal choices

[Step 2] osp_optimize

-> optimized ['Lipophilicity', 'Permeability', 'Intrinsic clearance', 'Specific intestinal permeability (transcellular)'] on 4 study(ies) -> GMFE 1.524 (best so far 1.524)

optimized={'Lipophilicity': 1.4920107663657072, 'Permeability': 0.00037520157796026095, 'Intrinsic clearance': 0.41670803288507974, 'Specific intestinal permeability (transcellular)': 3.8376090327163534e-05}

[Step 3] osp_optimize

reason: Good result already (GMFE 1.524, meets ≤ 1.6 target). Permeability and intestinal permeability have low sensitivity (weakly identified) but not at bounds - acceptable, since they still contribute plausibly to absorption kinetics. Let's check if fixing weakly-sensitive ones and refining fraction unbound (measured_soft) could improve fit, particularly cirrhosis IV underprediction and P0 overprediction bias. But since GMFE already ≤ 1.6 overall, this is sufficient. Let me do one more check - maybe fixing permeability params at reasonable value and re-running with fraction unbound freed within measured range to address bias asymmetry.

-> optimized ['Lipophilicity', 'Intrinsic clearance', 'Fraction unbound (plasma, reference value)'] on 4 study(ies) -> GMFE 1.609 (best so far 1.524) optimized={'Lipophilicity': 1.5189626753964576, 'Intrinsic clearance': 0.35266843869404396, 'Fraction unbound (plasma, reference value)': 0.10942782601812169}

[Step 4] osp_optimize

reason: This is worse than the first run. Revert to the original 4-parameter estimation (Stage 1), which gave GMFE 1.524, meeting the target.

-> optimized ['Lipophilicity', 'Permeability', 'Intrinsic clearance', 'Specific intestinal permeability (transcellular)'] on 4 study(ies) -> GMFE 1.524 (best so far 1.524)

optimized={'Lipophilicity': 1.4920107663657072, 'Permeability': 0.00037520157796026095, 'Intrinsic clearance': 0.41670803288507974, 'Specific intestinal permeability (transcellular)': 3.8376090327163534e-05}